

MIRZA TAIRIN TARIQUE

✉ mirza.tairin@gmail.com 🌐 mirzatairin in mirzatairin ☎ +880 1778 722 691 🏠 Dhaka, Bangladesh

ABOUT ME

Computer Science graduate and researcher with interests in Human-centered & Trustworthy AI.

EDUCATION

BRAC University Master of Science in Computer Science and Engineering	Fall 2026–
Ahsanullah University of Science & Technology Bachelor of Science in Computer Science & Engineering	2015 - 2019

WORK EXPERIENCE

Software Developer Global Spirit Engineering Trade & Services Ltd	2023 - 2025
Database Developer East Bengal Transport Service	2021 - 2022
Programmer, MIS Panjeree Publications Ltd	2019 - 2020

PUBLICATIONS

- The Good, the Bad, and the Potential of AI-based Systems in Computing Education
Accepted Paper in IEEE COMPSAC 2025: The ethical implications and human-centered perspectives of AI-assisted learning, as well as the challenges and opportunities of integrating AI-based systems into computer science higher education.

PRESENTATIONS & RESEARCH RECOGNITION

- **Beyond Answer Generation: Designing AI Companions for Independent Mathematical Problem Solving** (*GHC Academic Poster Sessions, Grace Hopper Celebration 2026, AWS-sponsored; October 2026*)
- **StormCore3D: Tracking Storms Over Time and Observing Patterns for Disaster Preparedness** (*Shortlisted Top Entry, IEEE SciVis Contest 2026; invited for in-person presentation and poster at IEEE VIS 2026*)
- **A Minimal Computing Approach to Community-Driven Preservation of Intangible Musical Heritage** (*Invited CFP University College Cork, Ongoing*)

TECHNICAL SKILLS

Programming & Engineering	Python, SQL, Java, C++, Django, .NET, Git
AI & Machine Learning	PyTorch, Scikit-learn, Pandas, NumPy, Computer Vision, Machine Learning, Fairlearn, LLM Fine-Tuning (QLoRA), Hugging Face
Research & Visualization	Matplotlib, Jupyter, ATLAS.ti, StreamLit

RESEARCH PROJECTS

- Gemma for Art (Fine-tuning Multimodal Vision-Language System for Artwork Analysis): Fine-tuned a multimodal vision-language model for art-historical reasoning, contextual interpretation, and structured analysis of visual artworks using QLoRA. (*Google DeepMind Hackathon, 2026*)
- Beyond Answer Generation: Designing AI Companions for Independent Mathematical Problem Solving: A human-AI learning framework investigating how AI systems can support independent mathematical problem solving rather than merely generate answers. (*GHC Academic Poster Sessions, Grace Hopper Celebration 2026, AWS-sponsored*)

- StormCore-3D: Thermodynamic Diagnostic Tool: Multi-field visual analytics framework for tracking and analyzing tropical cyclones across multi-terabyte, multi-resolution planetary simulation (ex. NASA's DYAMOND GEOS dataset). (*Shortlisted Top Entry, IEEE SciVis Contest 2026*)
- Automated License Plate Recognition (ALPR) System: End-to-end computer vision system for object detection and recognition under real-world environmental variability using a fine-tuned YOLOv8 pipeline (*Undergraduate Thesis*)
- Air Quality Analysis and AQI Modeling: Analyzed urban air quality dynamics using OpenAQ data for environmental modeling and data-driven decision-making. (*AWS AI/ML Scholars Program*)
- Studying Reasoning, Abstraction, and Generalization Across Humans and Intelligent Systems: Investigating reasoning, abstraction, hypothesis generation, and generalization across humans, large language models, and algorithmic systems using ARC as a research environment. (*ARC AGI-3 Paper Track, Ongoing*)
- Algorithmic Bias Mitigation in ML Models: Investigated fairness and bias mitigation in machine learning models, focusing on model reliability, decision transparency, and trustworthy AI.
- Exoplanet Atmosphere Spectral Analyzer: Computational framework simulating and analyzing near-infrared exoplanet atmospheric spectra. (*Planetary Science Project*)
- Digital Minds: Human-AI Coexistence Under Moral Uncertainty: A conditional policy framework designed for human-AI interaction under moral uncertainty. (*Apart Research, Submitted for Review*)
- Open Source Contributions: SHAP (SHapley Additive exPlanations): Contributed to an open-source explainable AI library for interpreting and auditing machine learning models.

TRAINING/CERTIFICATION

The Global Consumer Intelligence (GCI) World, Matsuo-Iwasawa Laboratory, The University of Tokyo	2026-ongoing
Future AWS AI Programmer Nanodegree Program	2026-ongoing
AWS AI Practitioner Challenge	2026
Introduction to User Experience Design (Georgia Institute of Technology)	2025

INTEREST & HOBBIES

Music Production, Creative Writing, Digital Art.

LANGUAGE

Bangla	Native
English	IELTS 8.0
Japanese	Beginner

REFERENCES

Dr. Mohammad Shafiu Alam
 Professor
 Computer Science
 Ahsanullah University of Science & Technology
shafiu.cse@aust.edu

Dr. Moushumi Sharmin
 Professor
 Computer Science
 Western Washington University
sharmim@wwu.edu